

Stats exam

Question

1) understand which of the available variables are mostly related to the insurance claim, and give suggestions to the company on the profiles of the customers a marketing campaign should point at (2) evaluate if there is any significant difference between claims made by males rather than by females, given the effect of the other factors (3) evaluate which region (among the 4) is more profitable for the company, while taking into account the effect of the other factors.

Required libraries used

```
library(alr4)
```

```
library(readr)
```

```
library(boot)
```

```
library(readxl)
```

I have imported the csv file by converting into excel to csv as it's not readable in excel.

Then I can see some NA values in my analysis so I used `na.omit` function to remove all the NA values.

Then I have used `str` function to read my data.

So that I can find that I have some categorical values in it.

Then I have used the `as.factor` function to make them as factors

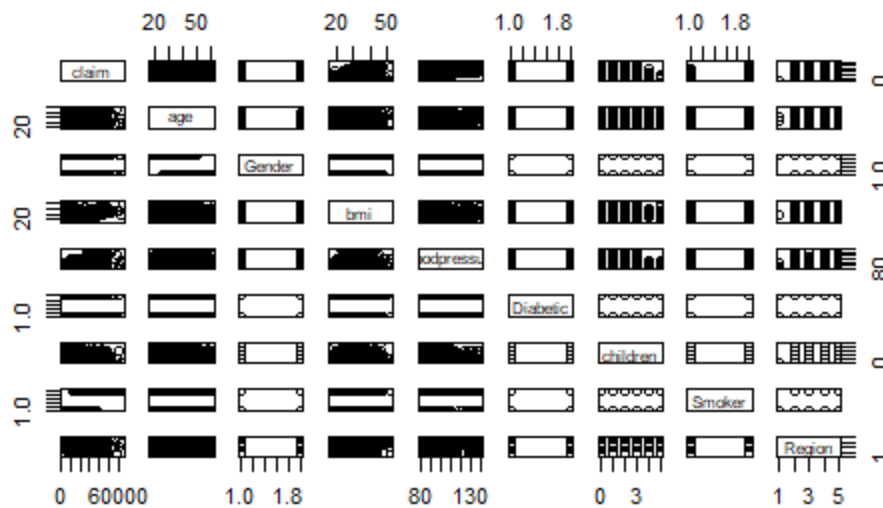
```
Gender=as.factor(m1$gender)
```

```
Region=as.factor(m1$region)
```

```
Smoker=as.factor(m1$smoker)
```

```
Diabetic=as.factor(m1$diabetic)
```

Then I have done the pairs to see the marginal plots distribution.



According to the marginal plots I can see that there is a linear distribution between bmi and claim and bloodpressure and claim and I can see some outliers and the variability is fine with normal distribution.

Then I have selected the specific variables and checked for vif and I don't see any collinearity and I used lm function

Then my next step is to do residual plots in the initial I had quadratic in my plots by rejecting the null hypothesis but I have done some transformations.

This is the model

$M2 = \text{lm}(I(\text{claim}^{0.265}) \sim \text{age} + \text{bmi} + \text{Gender} + \text{bloodpressure} + I(\text{bloodpressure}^2) + \text{Diabetic} + \text{children} + I(\text{children}^2) + \text{Smoker} + \text{Region}, \text{CLAIM})$

The residual plot data is

	Test stat	Pr(> Test stat)
age	-0.5120	0.60875
bmi	1.6816	0.09288 .
$I(\text{bmi}^2)$	-1.1515	0.24974
Gender		
bloodpressure	1.3584	0.17457
Diabetic		

children -1.7042 0.08857 .

Smoker

Region

Tukey test -0.9710 0.33152

I can see that the p-value shows not to reject the hypothesis by accepting the alternative hypothesis, where there is no quadratic in plots, which is greater than 0.05.

Then I have did the ncv test and I didn't get any value which is less than 0.05 which shows we do not reject null hypothesis then I have seen the outliers test and it doesn't affect the data because the Bonferroni p-value is more than 0.05.

Non-constant Variance Score Test

Variance formula: ~ fitted.values

Chi-square = 25.37431, Df = 1, p = 4.721

No Studentized residuals with Bonferroni $p < 0.05$

Largest |rstudent|:

rstudent unadjusted p-value Bonferroni p

1161 3.382753 0.00073863 0.98607

Call:

lm(formula = l(claim^0.352) ~ age + bmi + l(bmi^2) + Gender +
bloodpressure + Diabetic + children + Smoker + Region, data = m1)

Residuals:

Min	1Q	Median	3Q	Max
-12.2642	-3.5972	-0.3077	3.2507	16.7713

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-16.467684	4.170234	-3.949	8.27e-05 ***

age	0.013530	0.013447	1.006	0.314504	
bmi	0.809776	0.172092	4.705	2.80e-06	***
l(bmi^2)	-0.009723	0.002705	-3.594	0.000337	***
Gendermale	-0.340740	0.299252	-1.139	0.255060	
bloodpressure	0.151817	0.013387	11.341	< 2e-16	***
DiabeticYes	-0.180703	0.274589	-0.658	0.510596	
children	0.832887	0.113762	7.321	4.25e-13	***
SmokerYes	13.121913	0.376158	34.884	< 2e-16	***
Regionnortheast	11.161642	2.910849	3.834	0.000132	***
Regionnorthwest	8.715043	2.903423	3.002	0.002736	**
Regionsoutheast	8.050866	2.903204	2.773	0.005631	**
Regionsouthwest	8.789971	2.906037	3.025	0.002537	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.993 on 1322 degrees of freedom

Multiple R-squared: 0.6409, Adjusted R-squared: 0.6376

F-statistic: 196.6 on 12 and 1322 DF, p-value: < 2.2e-16

This is the final fitted values and I can see the region northeast is impacting the claim more comparing to the other regions by 11.16. so 1 unit increase in northeast region impacts the claim by 11.16 compared with other variables.

By summary I can say almost all the variables impact the claim other than diabetic so that marketing campaign should make an eye on the diabetic customers. and according to the summary there p-value is more than 0.05 for male shows that female has a good impact but male doesn't have any impact on claim with respect to the other variable.

